Project Report

Comprehensive Analysis and Dietary Strategies with Tableau: A College Food Choices Case Study

Role	Name
Team Lead	Rohith Sai Kiran Patnala
Team Member	Shaik Sabiha
Team Member	Cheekatla Uday Sai Bhaskar

1. Introduction

1.1 Project Overview

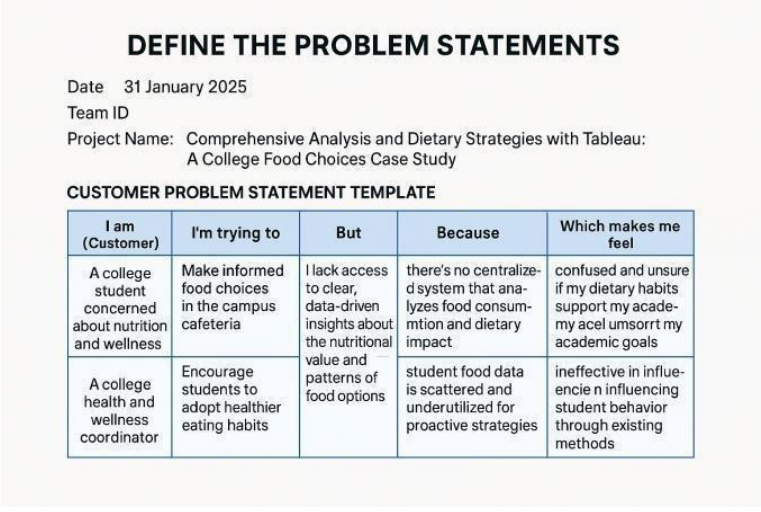
This project collects and analyzes food consumption data from students to identify dietary habits. Using Tableau dashboards, it presents clear and interactive visuals to support informed decision-making for both students and health coordinators. The goal is to promote nutrition awareness and improve wellness on college campuses.

1.2 Purpose

The purpose of this project is to help college students make healthier food choices by analyzing their eating patterns using Tableau visualizations.

2. Ideation Phase

2.1 Problem Statement



DEFINE THE PROBLEM STATEMENTS

Date 31 January 2025

Team ID

Project Name: Comprehensive Analysis and Dietary Strategies with Tableau: A College Food Choices Case Study

CUSTOMER PROBLEM STATEMENT TEMPLATE

I am (Customer)	I'm trying to	But	Because	Which makes me feel
A college student concerned about nutrition and wellness	Make informed food choices in the campus cafeteria	I lack access to clear, data-driven insights about the nutritional value and patterns of food options	there's no centralize- d system that ana- lyzes food consum- mtion and dietary impact	confused and unsure if my dietary habits support my acade- my acel umsorrt my academic goals
A college health and wellness coordinator	Encourage students to adopt healthier eating habits		student food data is scattered and underutilized for proactive strategies	ineffective in influe- encie n influencing student behavior through existing methods

Fig. 2.1 — Problem statement definition worksheet (Team ID, Project Name, Date: 31 January 2025)

PS No.	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	A college student with limited time,	Make healthy and affordable food choices on campus.	The available options are often unhealthy,	Campus dining prioritizes convenience over	Frustrated and stressed about balancing diet,

PS No.	I am (Customer)	I'm trying to	But	Because	Which makes me feel
	budget, and nutritional knowledge		expensive, or lack variety.	nutrition, and healthy options are priced higher.	academics, and finances.
PS-2	A campus dining administrator responsible for student wellbeing.	Improve student satisfaction and health outcomes through better food offerings.	Student feedback is scattered, and trends are unclear.	There's no centralized tool to analyze food choice data effectively.	Overwhelmed and unsure how to make data-driven decisions.

2.2 Empathy Map Canvas

Three scenarios were mapped to understand how different personas would experience and act on the food-choice data:

- Scenario 1 — Monitoring Nutritional Intake: a student or coordinator receives an alert flagging a concerning trend in nutritional intake, such as a significant drop in fruit and vegetable consumption.
- Scenario 2 — Addressing Dietary Deficiencies: predictive analytics are used to identify widespread dietary deficiencies, giving early, data-driven access to college food-choice patterns.
- Scenario 3 — Predictive Analysis and Personalized Nutrition Plans: predictive analytics capabilities are used to anticipate and help prevent unhealthy eating habits before they become established.

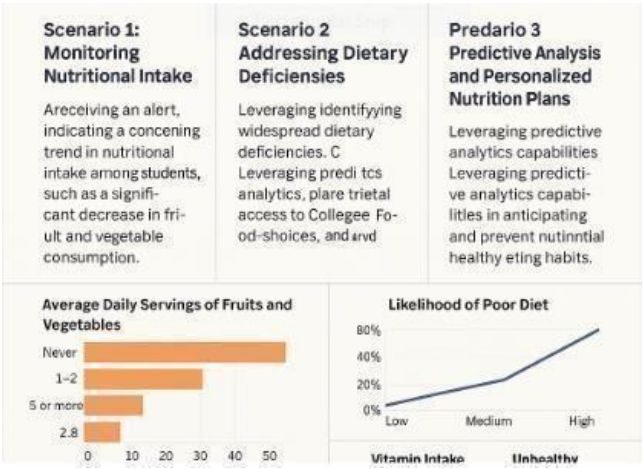


Fig. 2.2a — Average Daily Servings of Fruits and Vegetables (most students report eating fruits/vegetables rarely; "Never" is the largest group)

A companion chart in the same worksheet, "Likelihood of Poor Diet," plots poor-diet likelihood (0–80%) against vitamin intake level (Low / Medium / High), showing that poor-diet likelihood rises sharply as vitamin intake moves from Low to High-unhealthy patterns — i.e., students with erratic vitamin intake are disproportionately likely to also have a poor overall diet.

2.3 Brainstorming

Idea	Idea Description	Group / Category
1	Analyze meal frequency (breakfast/lunch/dinner) patterns	Meal Timing
2	Compare healthy vs unhealthy food choices across genders	Gender Comparison
3	Identify departments with the highest junk food consumption	Departmental Insights
4	Study stress levels and their relation to eating habits	Behavioral Impact
5	Use heat maps to show food preference intensity by hostel/region	Visualization Techniques

Idea	Idea Description	Group / Category
6	Embed the interactive dashboard in a Flask-based web app	Deployment / Web Integration
7	Create a Tableau story for dietary habit transformation journeys	Narrative & Communication
8	Use filters for department, gender, and BMI in the dashboard	Dashboard Interactivity
9	Show seasonal or exam-time food intake trends	Seasonal / Academic Stress Patterns
10	Add calculated fields for daily calorie or nutrient totals	Data Processing / KPIs
11	Forecast future obesity or malnutrition risks using data	Predictive Analysis
12	Show total and average sugar/salt intake across students	KPI Overview / Health Indicators

3. Requirement Analysis

3.1 Customer Journey Map



Fig. 3.1 — Customer journey map across Awareness, Consideration, and Decision stages for Persona 1 (College Student) and Persona 2 (Campus Health Coordinator)

Persona 1 (College Student): notices fatigue and poor focus, searches for nutrition information, and ultimately adopts Tableau-based food-choice visualizations to make more informed decisions.

Persona 2 (Campus Health Coordinator): notices an increase in student health complaints, researches tools for tracking and improving food offerings, and partners with the project team to use Tableau for institutional insight.

3.2 Solution Requirement

Functional Requirements

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	Registration through Form Registration through Gmail Registration through LinkedIn
FR-2	User Confirmation	Confirmation via Email Confirmation via OTP
FR-3	Data Upload & Storage	Upload CSV File Store into MySQL Database
FR-4	Data Visualization	Create visualizations in Tableau Integrate with dashboard Create time-, region-, and sector-based charts
FR-5	Dashboard Access	View interactive Tableau dashboard Use filters (Year, Region)
FR-6	Web Integration	Embed Tableau dashboard into Flask-based UI

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-7	Insights & Reports	View data stories Access summary reports on usage patterns

Non-Functional Requirements

NFR No.	Requirement	Description
NFR-1	Usability	The interface should be intuitive and user-friendly for both technical and non-technical users.
NFR-2	Security	Secure login with OTP/Email verification, and protected access to student data.
NFR-3	Reliability	The system should consistently provide correct, accurate visualizations.
NFR-4	Performance	Dashboards should load within 3–5 seconds even for large datasets.
NFR-5	Availability	The platform should be available 24/7 without major downtime.
NFR-6	Scalability	The system should support the addition of new datasets and new visualizations without redesign.

3.3 Data Flow Diagram

The data flow follows a linear pipeline from data collection through to archiving, with a review loop that can send data back to earlier steps if needed:

Step	Process
1	Start
2	Data Collected — dietary and food-choice survey data is gathered from the source dataset
3	Decision: is data sourced from the Kaggle food-choices dataset?
4	Cleaned & Preprocessed
5	Calculated Fields Added
6	Visualization Created
7	Decision: build charts in Tableau
8	Multiple worksheets created for different insights
9	Dashboard Published
10	Reviewed by Users
11	Decision: can this lead to revisiting earlier steps if needed? (loops back to step 4 if yes)
12	Archived
13	End

3.4 Technology Stack

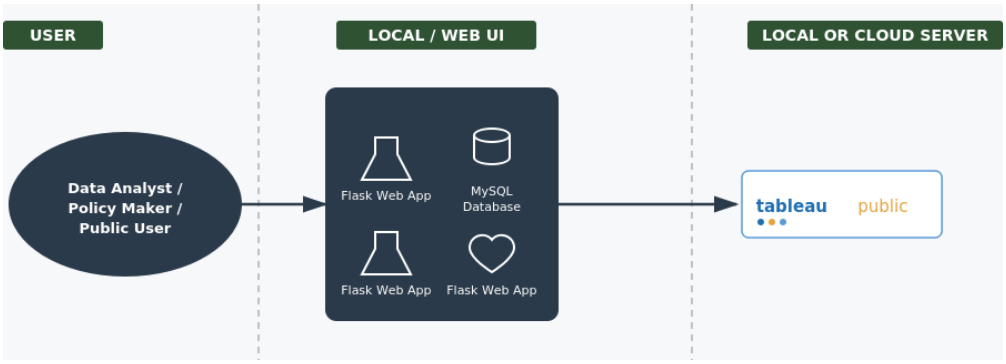


Fig. 3.4 — Technology stack: Data Analyst / Policy Maker / Public User accesses a Flask web app backed by a MySQL database; the app publishes visualizations to Tableau Public.

4. Project Design

4.1 Problem–Solution Fit

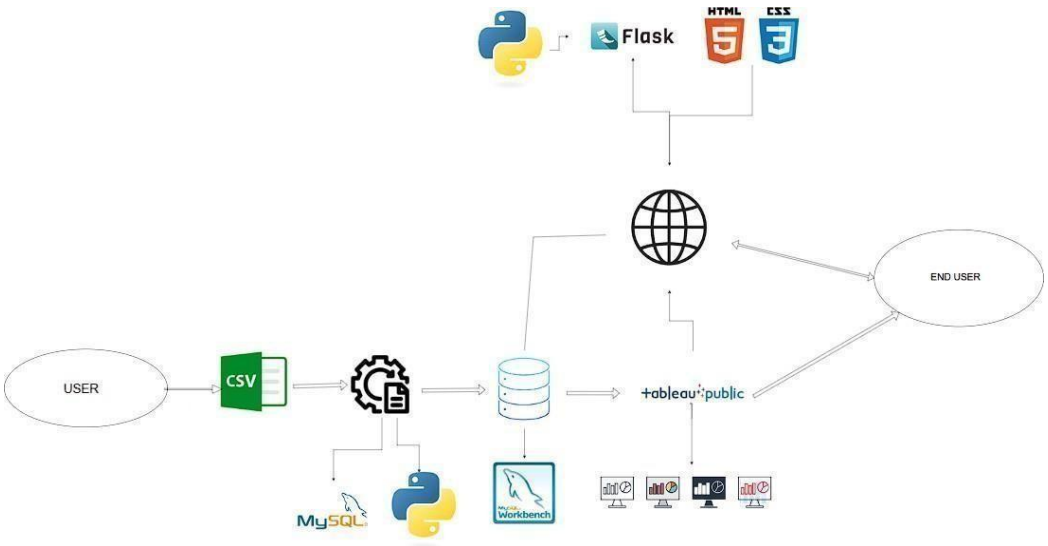


Fig. 4.1 — Problem–Solution Fit canvas covering Key Stakeholders, Current Challenges, Project Triggers, Use Case Scenarios, Tableau Solutions, Expected Benefits, Key Data Insights, Target Outcomes, and Implementation Strategy

4.2 Proposed Solution

S.No.	Parameter	Description
1	Problem Statement	College students often struggle to make healthy food choices due to limited nutritional awareness, budget constraints, and lack of accessible dietary data. This leads to poor eating habits, health issues, and decreased academic performance.
2	Idea / Solution Description	This project analyzes the dietary patterns of college students using data visualization with Tableau. By collecting data from surveys and food consumption logs, the system provides personalized nutritional insights, visual patterns, and suggested improvements.
3	Novelty / Uniqueness	The uniqueness lies in integrating dietary analysis with intuitive data visualization tools like Tableau specifically for a college demographic. Unlike generic health apps, this approach focuses on academic environments, combining nutrition, affordability, and accessibility.
4	Social Impact / Customer Satisfaction	This solution promotes better health among college students, potentially reducing stress and improving focus, energy levels, and academic outcomes. It encourages healthy living habits at a critical life stage.
5	Business Model (Revenue Model)	The project can adopt a freemium model where basic visual insights are free, while advanced features (detailed dietary tracking, custom meal plans, cafeteria integration) are offered as premium services.
6	Scalability of the Solution	The model is scalable across institutions by adapting data input formats and user preferences, and can expand to include meal-logging integrations, cross-campus comparisons, and regional health initiatives.

4.3 Solution Architecture

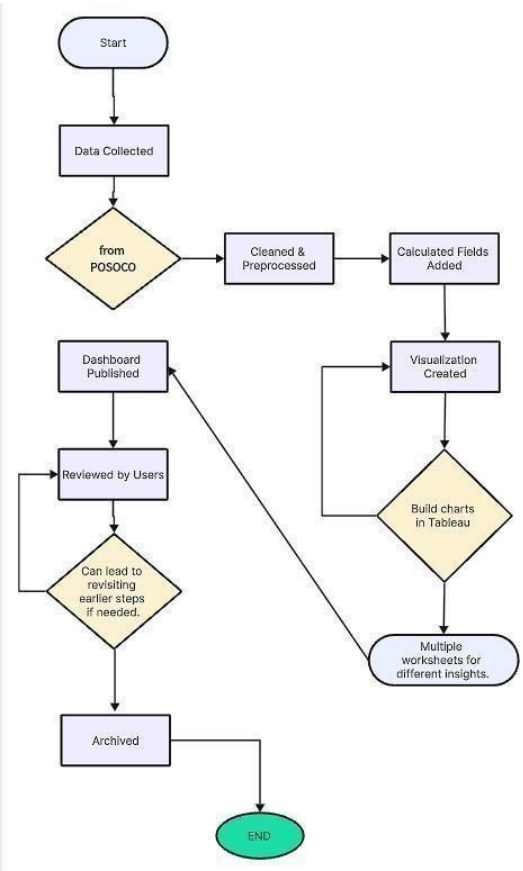


Fig. 4.3 — Solution architecture: data flows from the user (CSV upload) through preprocessing into MySQL, is visualized in Tableau Public, and served to the end user through a Python/Flask web app.

The backend is built with Python and Flask, which handles CSV upload, validation, and preprocessing. Cleaned data is stored in a MySQL database, and connected to Tableau, where the dashboard and story visualizations are built and published to Tableau Public. The Flask front end (HTML5/CSS3) embeds the published Tableau views so that end users — students, health coordinators, or the public — can interact with filters and read insights without needing a Tableau license of their own.

5. Project Planning & Scheduling

Sprint	Epic	User Story	User Story / Task	Points	Priority	Assigned To
Sprint-1	Registration	USN-1	As a user, I can register with my name and email	2	High	Rohith Sai Kiran Patnala
Sprint-1	Upload CSV	USN-2	As a user, I can upload dietary data in CSV format	3	High	Cheekatla Uday Sai Bhaskar
Sprint-1	Data Cleaning	USN-3	As a developer, I can clean and preprocess uploaded data using Python	4	High	Shaik Sabiha
Sprint-1	Database Storage	USN-4	As a developer, I can store cleaned data into MySQL	2	Low	Shaik Sabiha
Sprint-2	Tableau Dashboard	USN-5	As a user, I can view dashboards generated using Tableau	5	High	Rohith Sai Kiran Patnala
Sprint-2	Web Integration	USN-6	As a user, I can access the dashboard via Flask UI	3	High	Shaik Sabiha
Sprint-2	Add Filters	USN-7	As a user, I can filter the data by GPA, breakfast, and drink	2	Medium	Shaik Sabiha
Sprint-3	Data Story	USN-8	As a user, I can view a Tableau Story with key food diet insights	2	Low	Cheekatla Uday Sai Bhaskar
Sprint-3	Forecasting	USN-9	As a developer, I can forecast usage using Prophet	3	Low	Rohith Sai Kiran Patnala
Sprint-3	Documentation	USN-10	As a team, we can prepare final project documentation	2	Medium	Shaik Sabiha
Sprint-4	Deployment	USN-11	As a developer, I can deploy the Flask app and publish the Tableau dashboard online	3	High	Cheekatla Uday Sai Bhaskar
Sprint-4	Demo Prep	USN-12	As a team, we can prepare a live demo walkthrough for stakeholders	2	Medium	Rohith Sai Kiran Patnala
Sprint-4	Bug Fixing	USN-13	As a developer, I can test and fix UI/visual bugs from user feedback	2	Medium	Cheekatla Uday Sai Bhaskar

6. Functional and Performance Testing

6.1 Performance Testing

1. Data Rendered

The data rendered in this project includes information on cafeteria food items such as names, ingredients, and nutritional values like calories, proteins, and fats. It also captures student food choices, analyzing daily and weekly meal patterns. Additional data includes consumption trends, such as the most and least preferred foods and peak dining times. Where available, basic health metrics like BMI and student feedback on energy levels or meal satisfaction are also considered. Demographic details such as age and gender may be used to identify trends across student groups. All of this data is visualized using interactive Tableau dashboards to provide insights into nutritional breakdowns, evolving food preferences, and personalized or group-based dietary recommendations.

2. Data Preprocessing

Data preprocessing involved cleaning and organizing raw data collected from cafeteria records and student food logs. Missing values were handled by either imputing averages or removing incomplete entries. Duplicates were eliminated to ensure accuracy, and food item names were standardized for consistency. Nutritional information was cross-verified and formatted into structured categories such as calories, proteins, fats, and carbohydrates. Date and time fields were converted into a consistent format to analyze food consumption trends over time. Finally, the cleaned dataset was transformed into a format suitable for Tableau visualization, enabling effective analysis and insights.

3. Utilization of Filters

- Date Filter — select a specific day, week, or month to view food consumption trends over time.
- Meal Type Filter — filter data by meal category (breakfast, lunch, dinner, snacks) for focused analysis.
- Food Category Filter — display specific food groups such as fruits, vegetables, fast food, and beverages.
- Calorie Range Filter — view meals within a specific calorie range (e.g., under 500 kcal).
- Protein/Fat/Carb Filter — nutrient-based filters to view foods by macronutrient composition.
- Student Group Filter — filter consumption patterns by student categories (e.g., year of study, gender).
- Satisfaction/Rating Filter — filter based on student feedback or ratings of meals, where collected.
- Time of Day Filter — analyze which foods are preferred during morning, afternoon, or evening hours.

4. Calculated Fields Used

Calculated Field Name	Formula (Tableau syntax)
All_Students	"All Students"
All_Comfort_Reasons	STR([comfort_food_reason]) + ", " + STR([reason2])
Calories Importance Level	IF [calories_day] > 2500 THEN "High" ELSEIF [calories_day] > 1800 THEN "Medium" ELSE "Low" END
Clean Comfort Reason	TRIM(LOWER([comfort_food_reason]))
Cleaned Exercise	IF [exercise] >= 3 THEN "Active" ELSE "Inactive" END
Comfort Food Reasons – Split 1	SPLIT([comfort_food_reason], ",", 1)
Comfort Food Reasons – Split 2	SPLIT([comfort_food_reason], ",", 2)
Comfort Food Source	IF CONTAINS(LOWER([comfort_food]), "home") THEN "Home" ELSE "Outside" END

Calculated Field Name	Formula (Tableau syntax)
Comfort Reason Combined	IF CONTAINS([comfort_food_reason], "stress") THEN "Stress" ELSE "Other" END
Constant	"1" or "All"
Cooking Frequency Label	IF [cook] >= 5 THEN "Frequent" ELSEIF [cook] >= 2 THEN "Occasional" ELSE "Rare" END
DietStatus	IF [diet_current_coded] = 1 THEN "Vegetarian" ELSE "Non-Vegetarian" END
Employment / Gender Label	IF [employment_coded] = 1 THEN "Employed" ELSE "Unemployed" END IF [gender] = 1 THEN "Male" ELSE "Female" END
GPA Category	IF [GPA] >= 3.5 THEN "High" ELSEIF [GPA] >= 2.5 THEN "Medium" ELSE "Low" END
gpa_numeric	FLOAT([GPA])
Marital Status	IF [marital_status] = 1 THEN "Single" ELSE "Married" END
Nutritional Check	IF [fruit_day] + [veggies_day] >= 4 THEN "Yes" ELSE "No" END
Record Count	1 (Tableau auto-generated field used for counting records)

5. Dashboard Design

Number of dashboards: 1

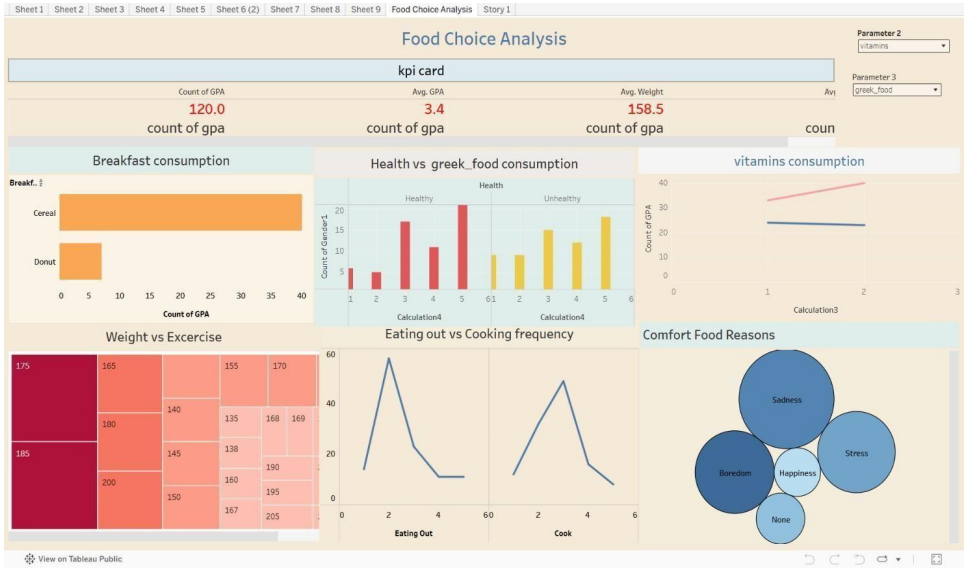
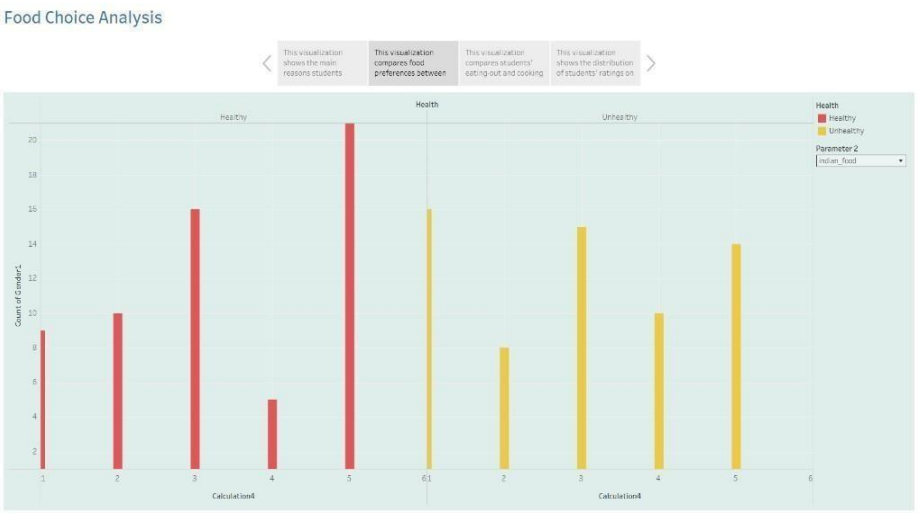


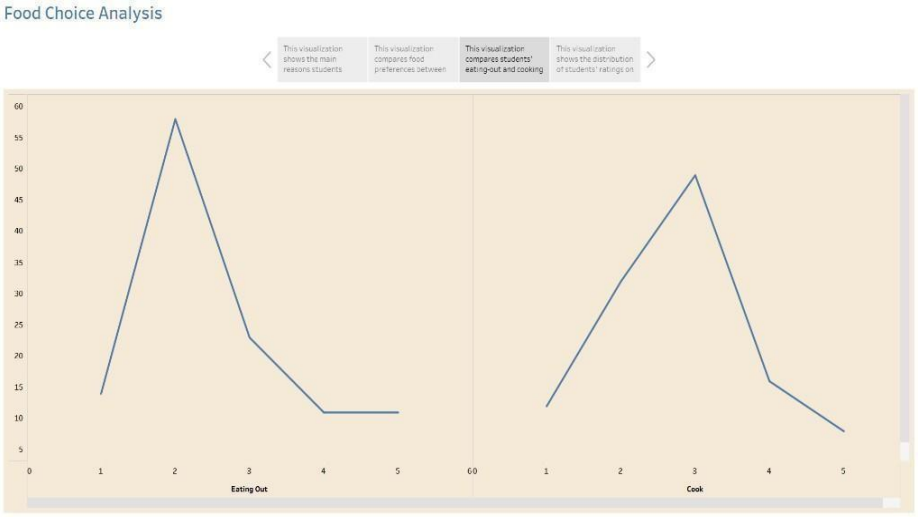
Fig. 6.1 — Dashboard 01 ("Food Choice Analysis"): KPI card, Breakfast Consumption, Health vs Greek-Food Consumption, Vitamins Consumption, Weight vs Exercise, Eating Out vs Cooking Frequency, and Comfort Food Reasons.

6. Story Designs

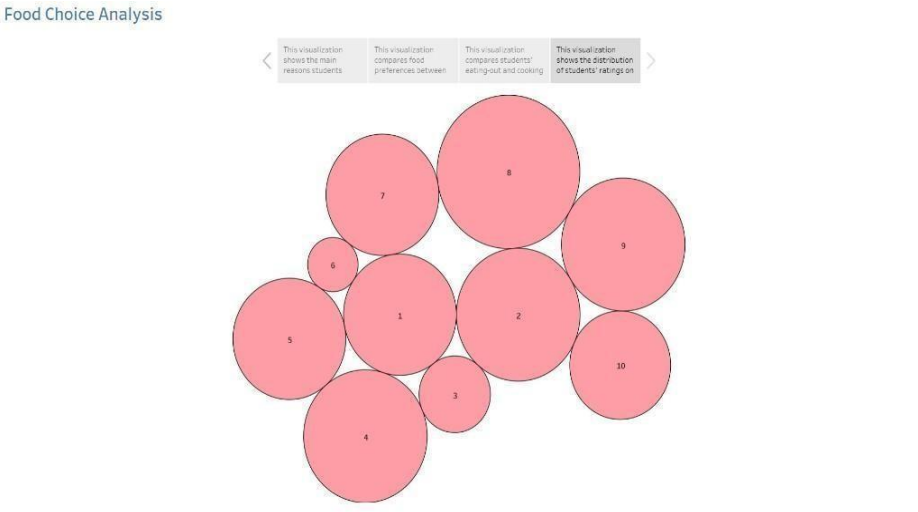
Number of stories: 1 story, made up of 8 visualization scenes.



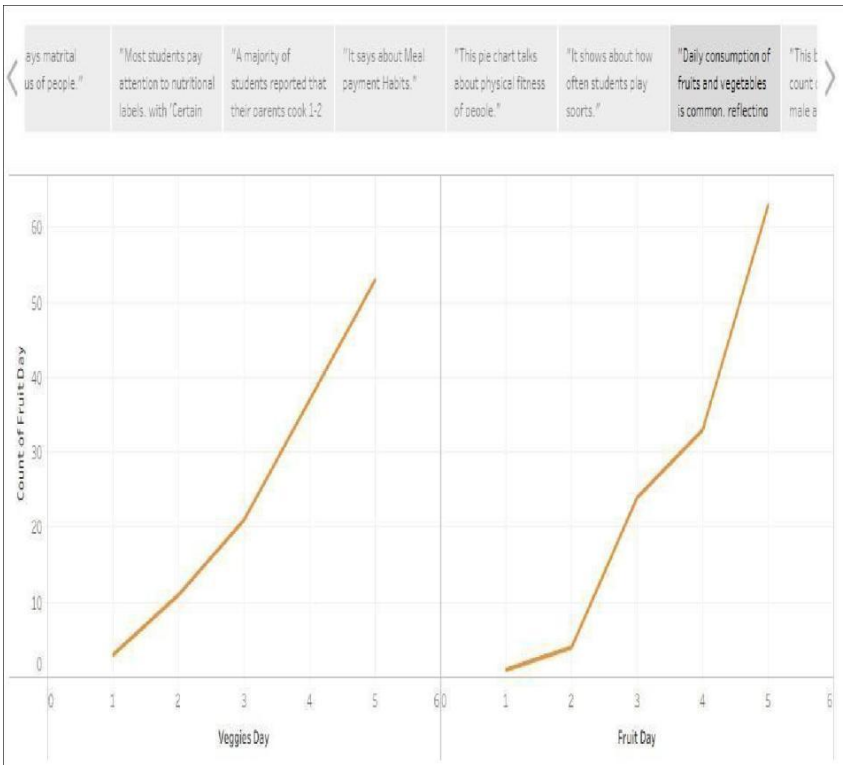
Opening view — the Food Choice Analysis dashboard that anchors the story.



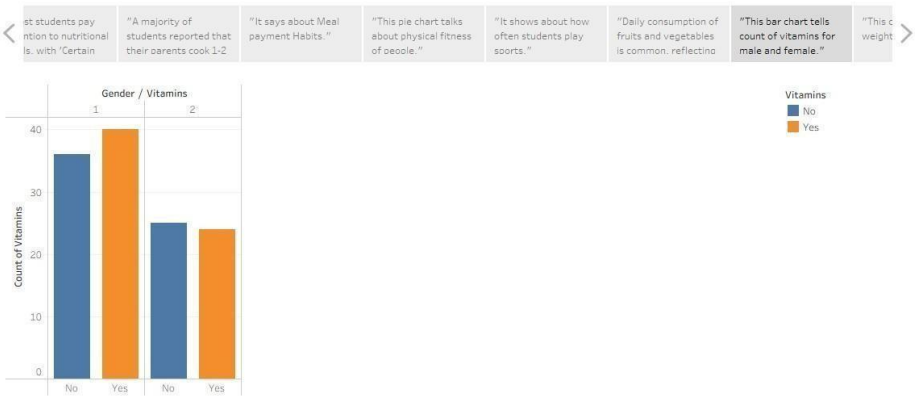
Scene 1 — main reasons students turn to comfort food (Sadness, Stress, Boredom, Happiness, None).



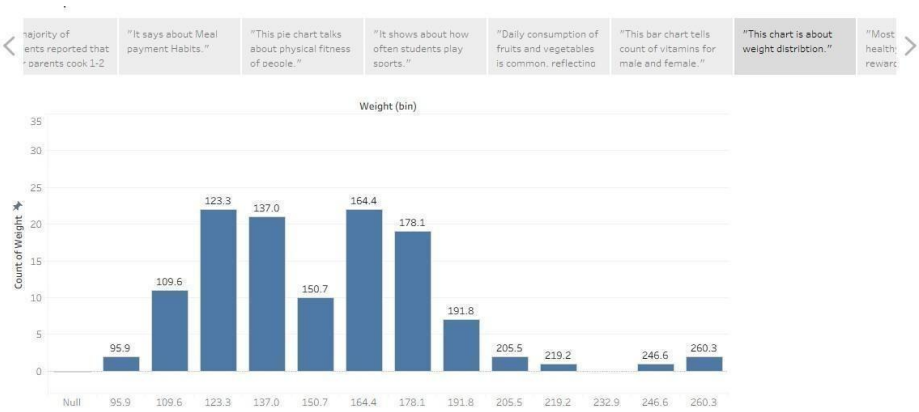
Scene 2 — comparison of students' food preferences between Healthy and Unhealthy categories.



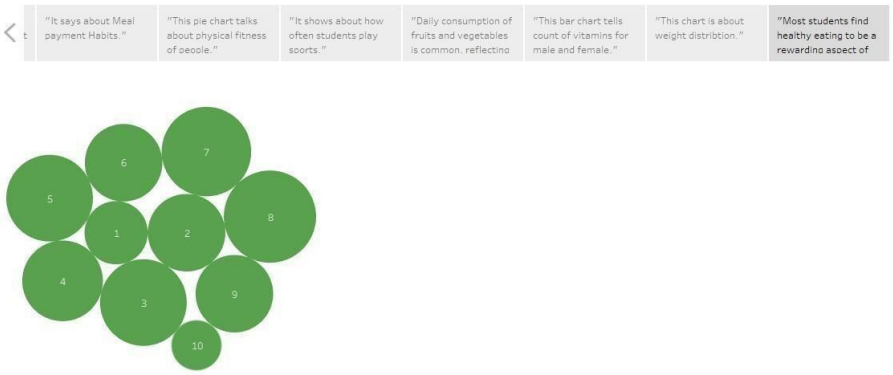
Scenes 3–4 — students' eating-out and cooking frequency, and the distribution of students' meal ratings.



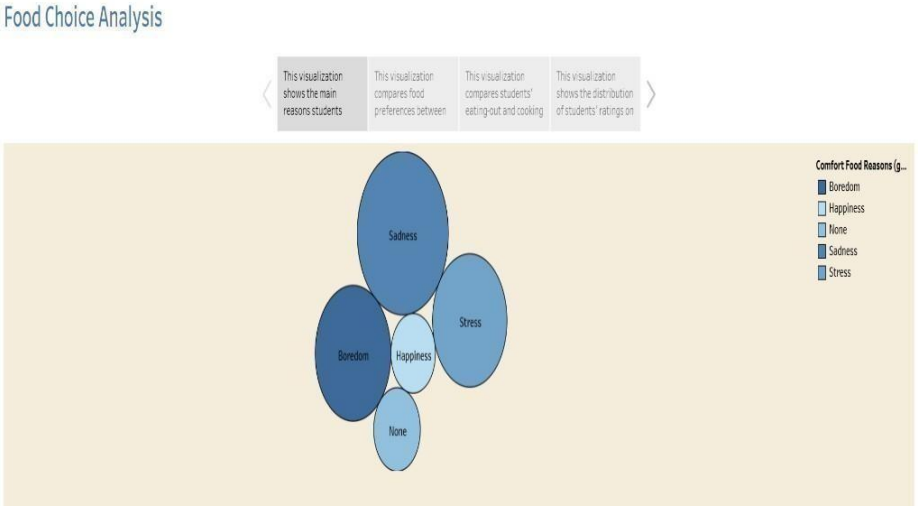
Scenes 5–6 — daily fruit and vegetable consumption trend, and vitamin intake split by gender.



Scene 7 — distribution of student weight.

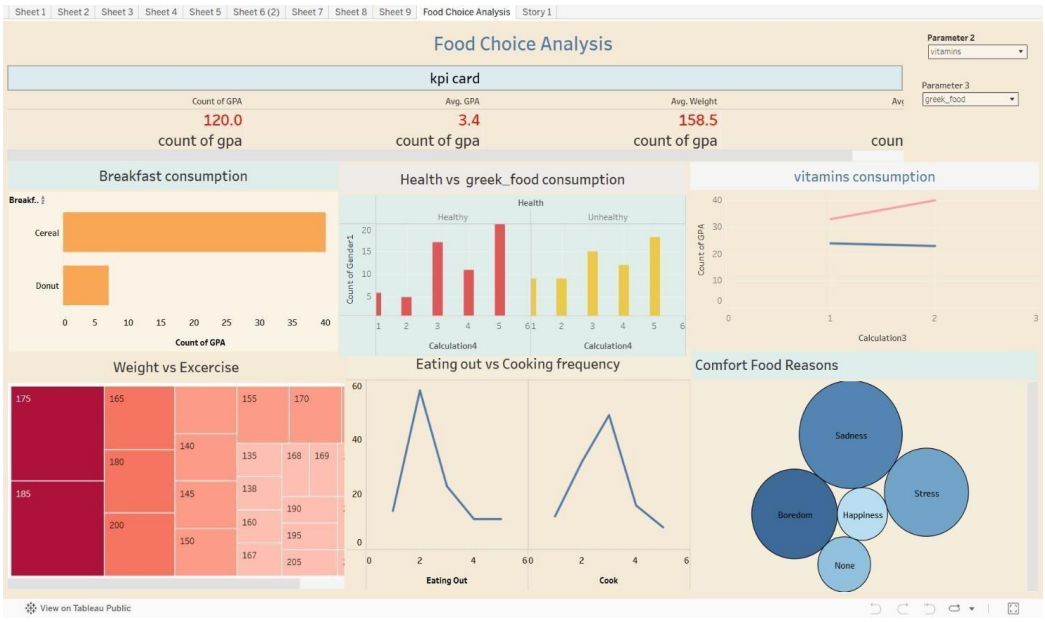


Scene 8 — most students find healthy eating to be a rewarding aspect of their routine.



7. Results

7.1 Output Screenshot



Dashboard 01 as published to Tableau Public (same dashboard shown in Fig. 6.1, Section 6).

The published dashboard and story confirm the trends found during analysis: most students report eating fruits and vegetables rarely, comfort eating is most often driven by sadness and stress rather than boredom or happiness, and reported vitamin intake and cooking frequency both skew toward the lower end of the scale. Health-vs-food-choice and weight-distribution views give health coordinators a quick way to see where nutritional gaps are concentrated across the student population, which was the core goal set out in the problem statement.

Dashboard link:

- https://public.tableau.com/shared/8M7NZRJ6C?:display_count=n&:origin=viz_share_link

Story links:

- https://public.tableau.com/shared/K24495CN3?:display_count=n&:origin=viz_share_link
- https://public.tableau.com/shared/75F3FXBXJ?:display_count=n&:origin=viz_share_link
- https://public.tableau.com/shared/5Q24J9KG2?:display_count=n&:origin=viz_share_link
- https://public.tableau.com/views/Food_Choice_Analysis1/Story1?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

8. Advantages & Disadvantages

Advantages	Disadvantages
Enables data-driven decision making for food choices	Data collection from cafeteria and students can be inconsistent
Promotes healthier eating habits among students	Potential privacy concerns with food consumption data
Uses Tableau for intuitive and interactive data visualization	Some users may resist using or learning the dashboard
Provides actionable insights to health coordinators and campus staff	Initial setup requires technical skill and effort

Advantages	Disadvantages
Supports personalized dietary suggestions	Requires ongoing maintenance and regular data updates
Educates students about nutrition and wellness	Limited impact if students do not engage with the tool
Scalable model for other educational institutions	Relies heavily on institutional support and collaboration

9. Conclusion

This project helps students choose healthier food using easy-to-understand Tableau dashboards. It supports better eating habits and wellness on campus. In the future, it can be improved and extended to more colleges.

10. Future Scope

- Personalized diet plans using AI/ML for student-specific recommendations
- Mobile app for real-time access and convenience
- Gamification to promote healthy eating habits
- Behavior tracking through long-term analytics
- Cafeteria insights for menu improvements
- Fitness app integration for a holistic health profile
- Admin dashboards for institutional health reporting
- Multi-campus expansion to replicate the model in other colleges
- Live feedback from students on meals
- Sustainability tracking via food waste and consumption data

Project Resources

- Source Code: Not published (NIL in submitted materials)
- Dataset: https://www.kaggle.com/datasets/borapajo/food-choices?select=food_coded.csv
- GitHub: https://github.com/Rohithsaikiran/college-food-choices-case-study
- Project Demo: https://drive.google.com/file/d/1NggkIe-0RduZ4cPWDDS_AytT-Zdtad-S/view?usp=sharing